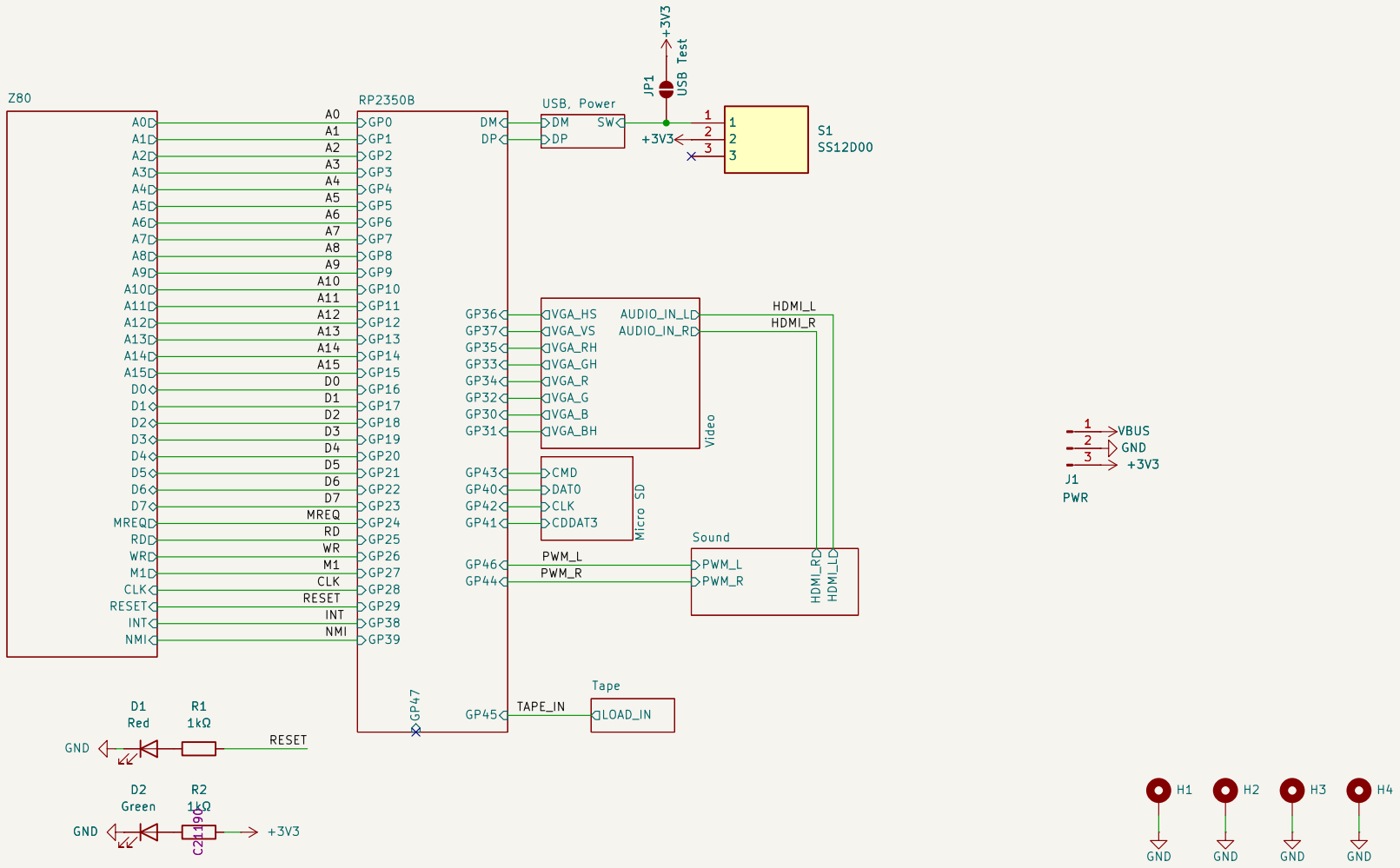


https://github.com/rh1tech/frank

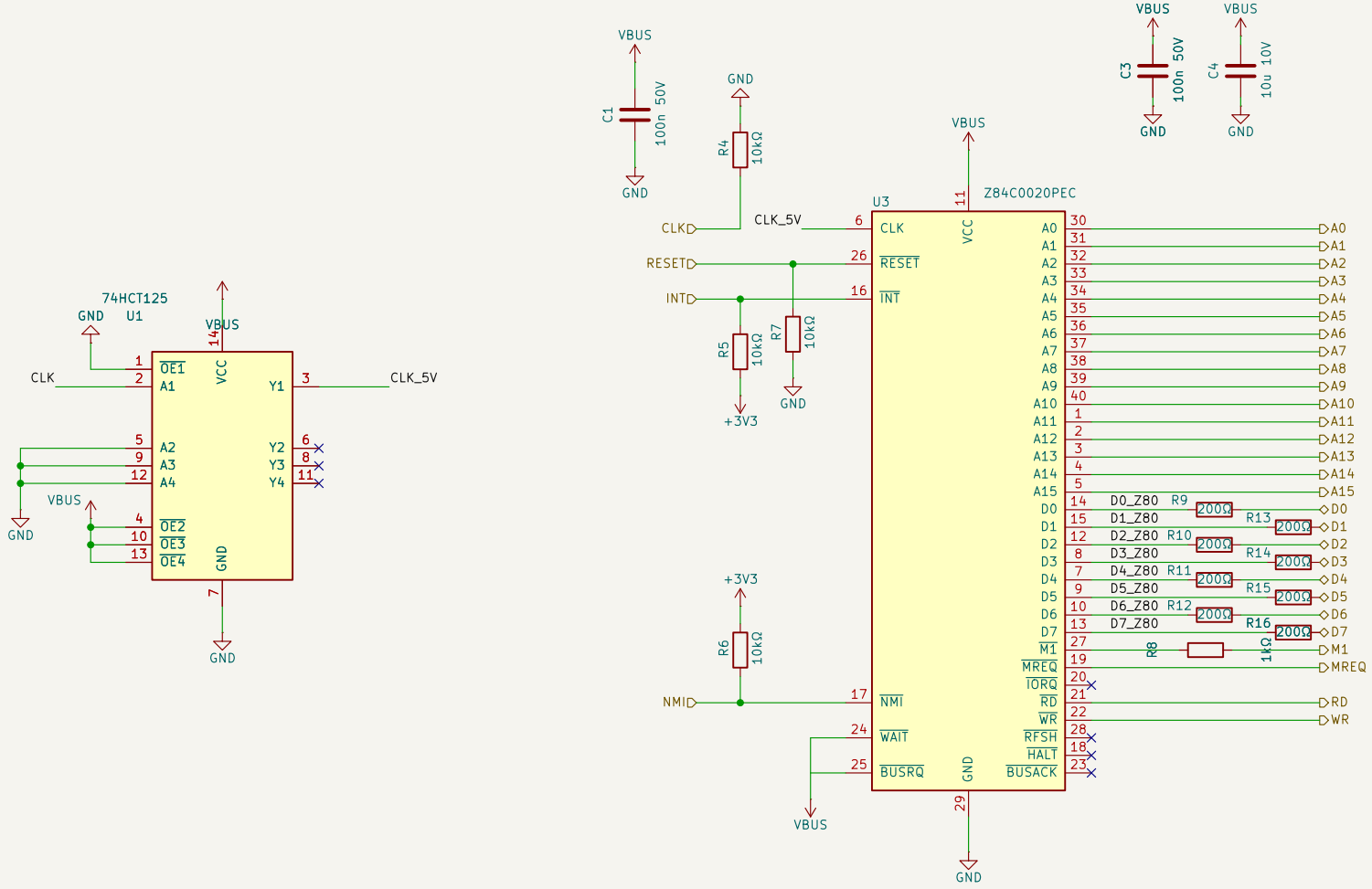
Mikhail Matveev

Sheet: /

File: dino_z80.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 1/16



https://github.com/rh1tech/frank

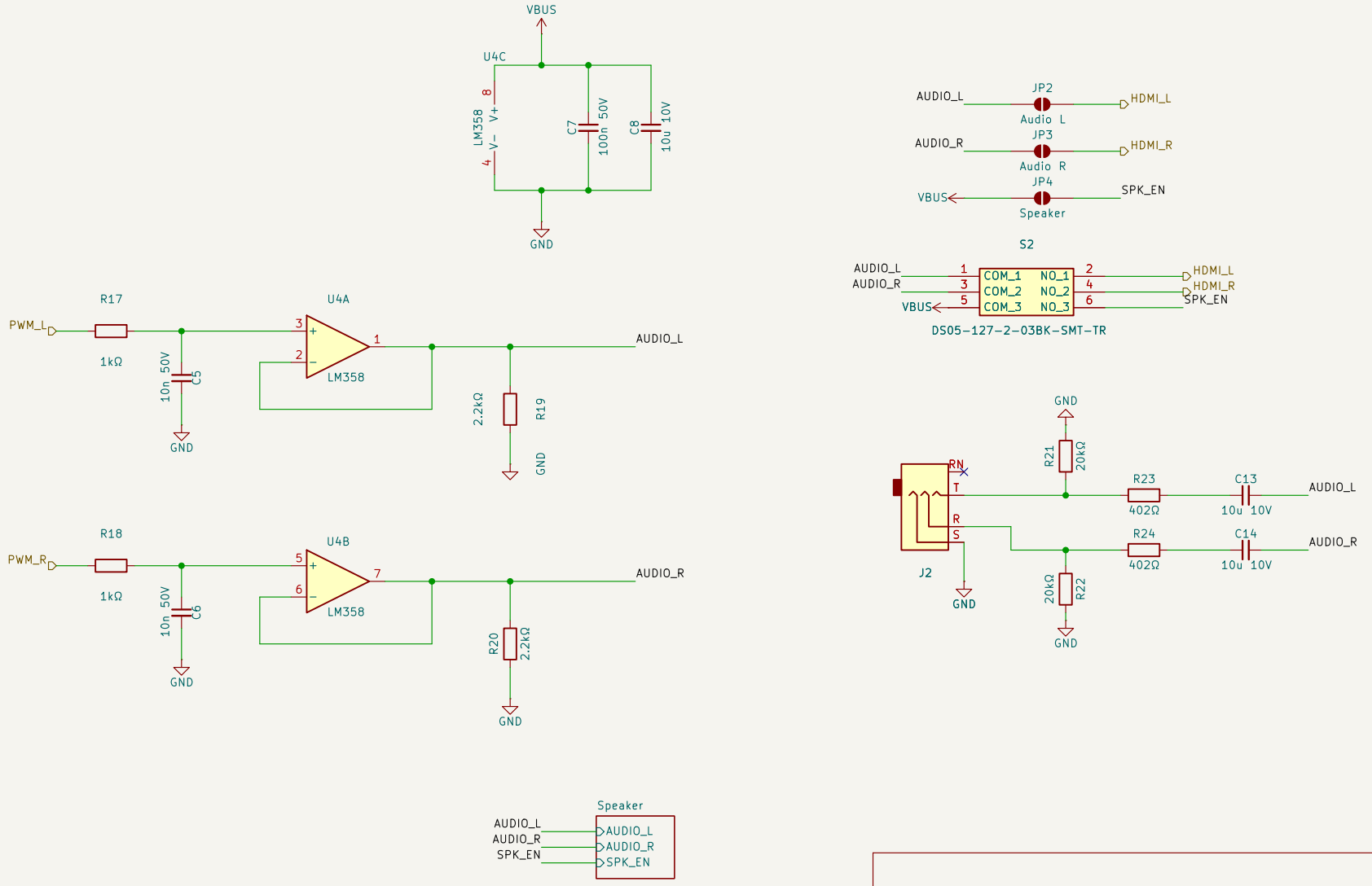
Mikhail Matveev

Sheet: /Z80/

File: z80.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 2/16



https://github.com/rh1tech/frank

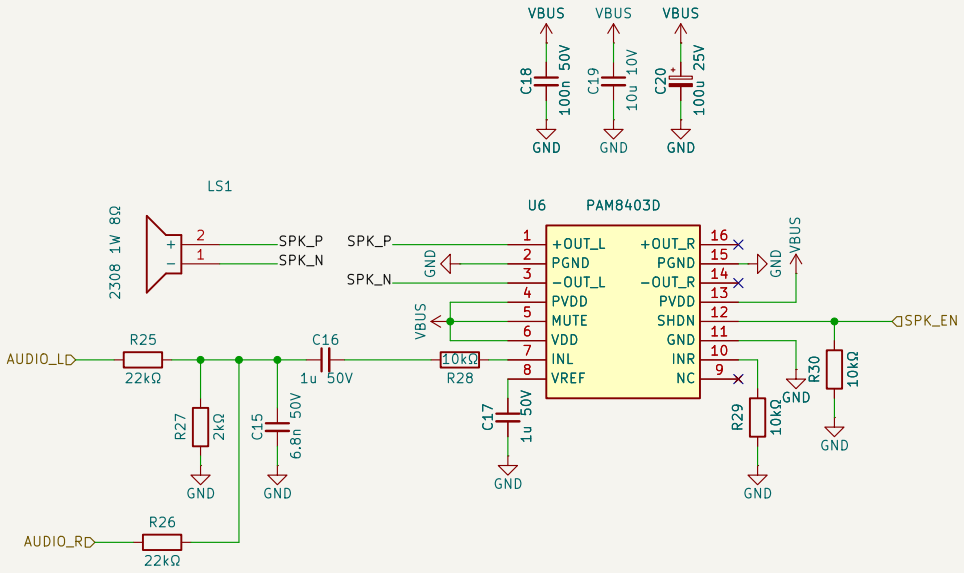
Mikhail Matveev

Sheet: /Sound/

File: sound.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 3/16



https://github.com/rh1tech/frank

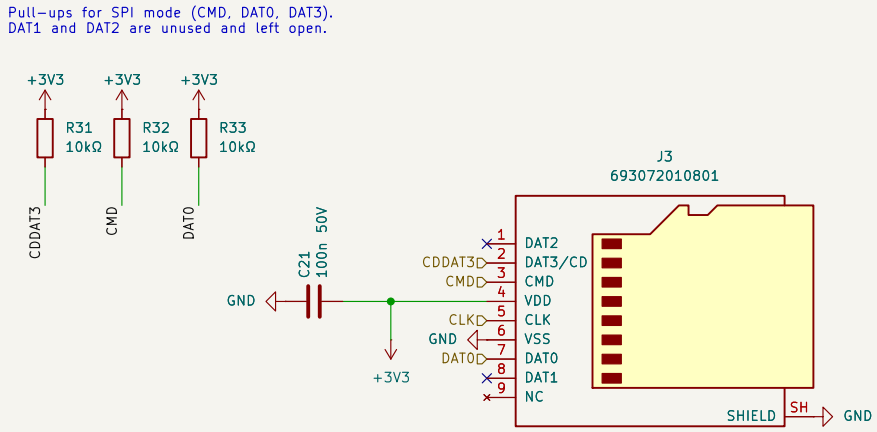
Mikhail Matveev

Sheet: /Sound/Speaker/

File: speaker.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 4/16



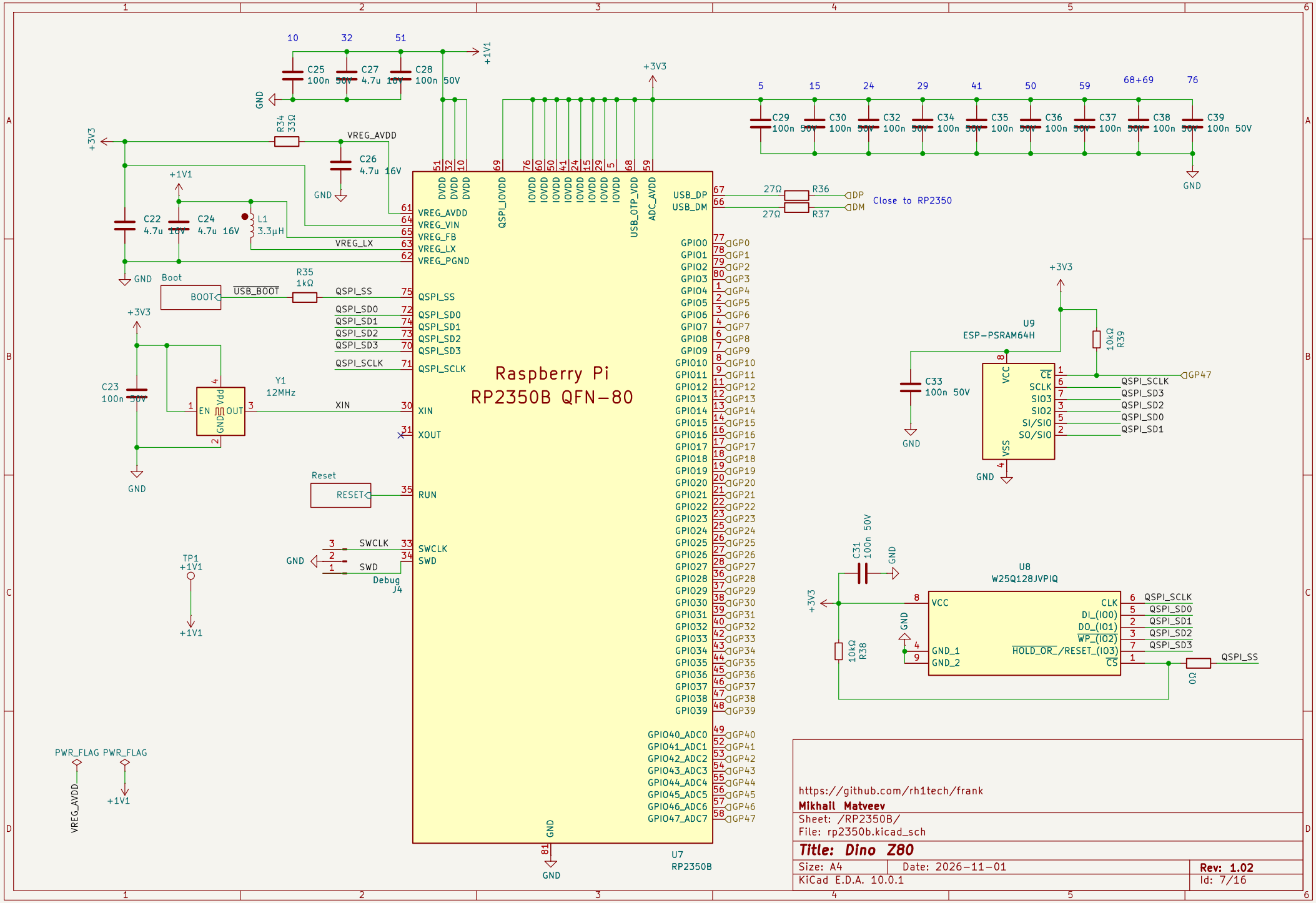
https://github.com/rh1tech/frank

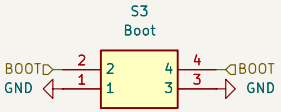
Mikhail Matveev

Sheet: /Micro SD/
File: sd.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 5/16


Raspberry Pi
RP2350B QFN-80
U7
RP2350B
U8
W25Q128JVPIQ
U9
ESP-PSRAM64H
Y1
12MHz
Boot
Reset
Debug
J4
TP1
Close to RP2350
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /RP2350B/
File: rp2350b.kicad_sch
Title: Dino Z80
Size: A4
Date: 2026-11-01
Rev: 1.02
KiCad E.D.A. 10.0.1
Id: 7/16



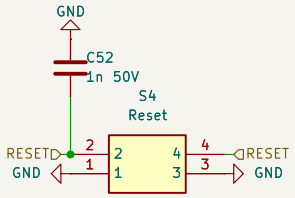
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /RP2350B/Boot/
File: boot.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 4/16



https://github.com/rh1tech/frank

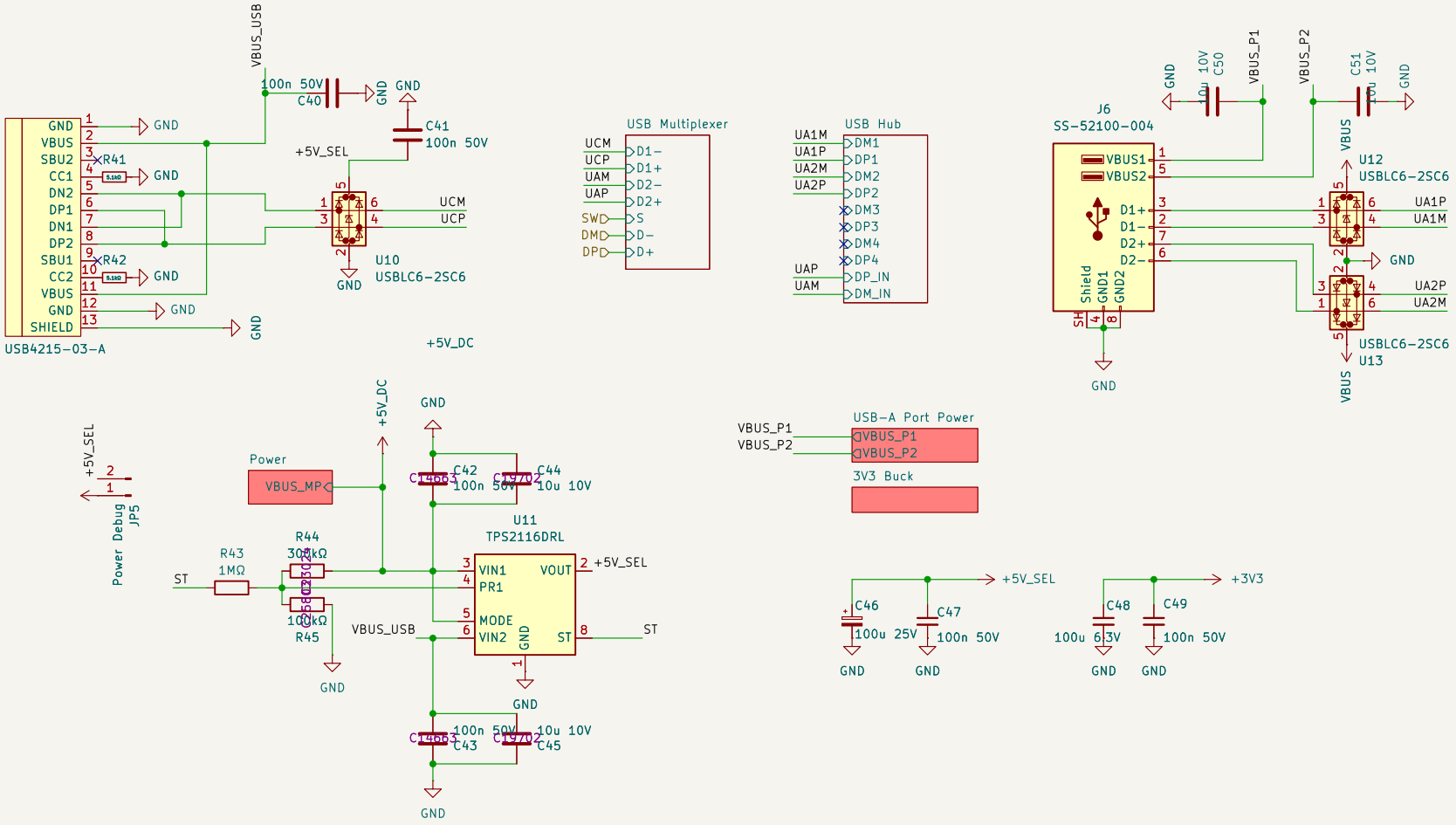
Mikhail Matveev

Sheet: /RP2350B/Reset/

File: reset.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 9/16



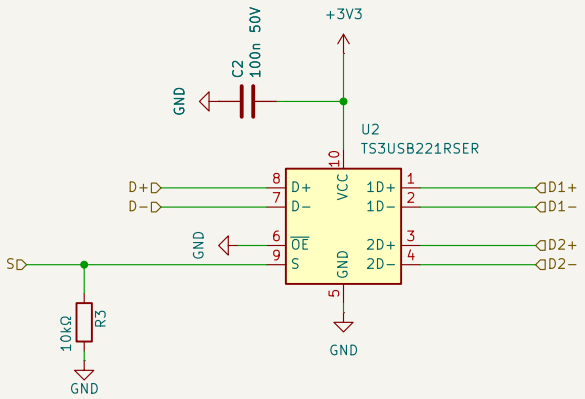
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /USB, Power/
File: usb.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 8/16



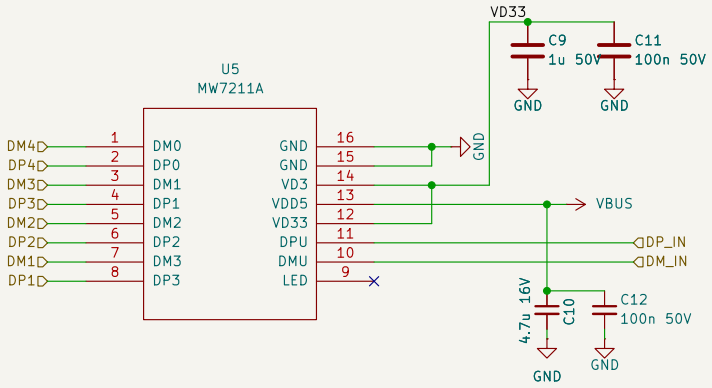
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /USB, Power/USB Multiplexer/
File: usb_multiplexer.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 2/16



https://github.com/rh1tech/frank

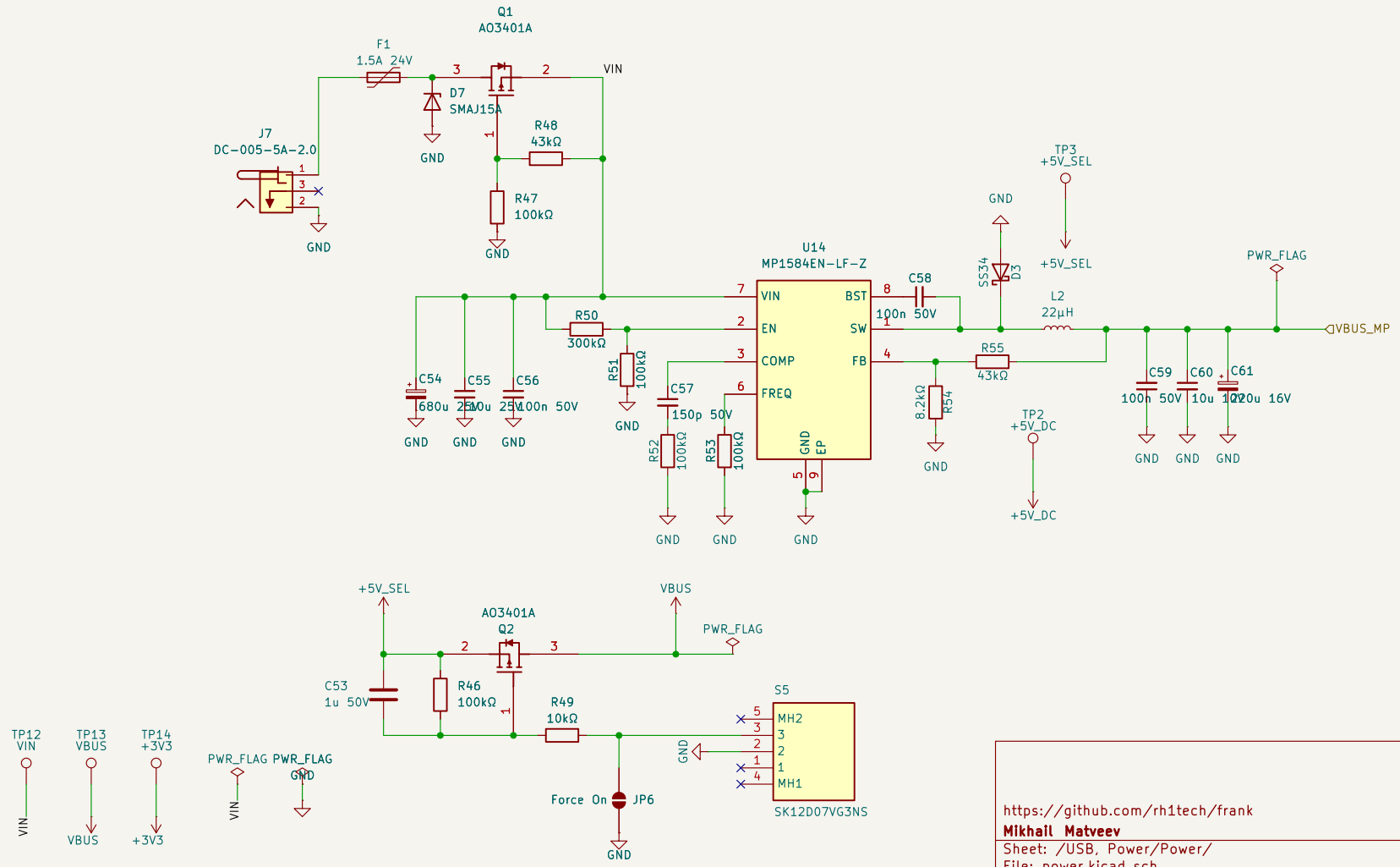
Mikhail Matveev

Sheet: /USB, Power/USB Hub/
File: usb_hub.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 3/16

DC input: 9–12 V (centre +), max 15 V: D7 SMAJ15A starts conducting at ~16.7 V and crowbars F1 on reverse polarity.
Q1 reverse-polarity P-FET: gate divider R47 100K / R48 43K keeps Vgs at −2.7 V (9 V) .. −7.3 V (TVS clamp).
MP1584 UVLO: R50 300K / R51 100K -> starts at ~6.0 V (6.6 V max). C54 680u 25V, C55 10u 25V.



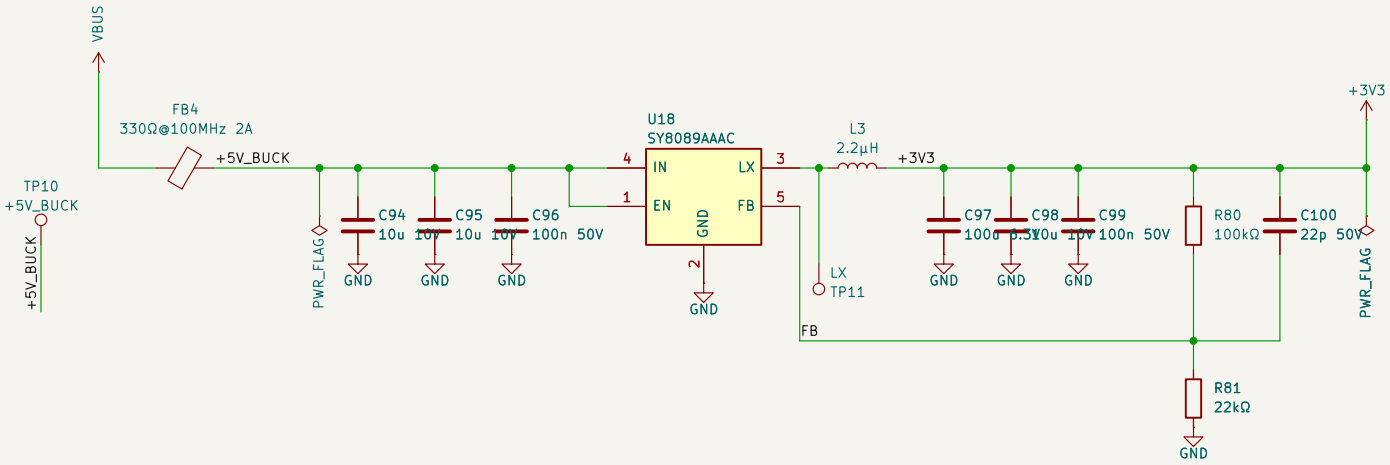
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /USB, Power/Power/
File: power.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 10/16



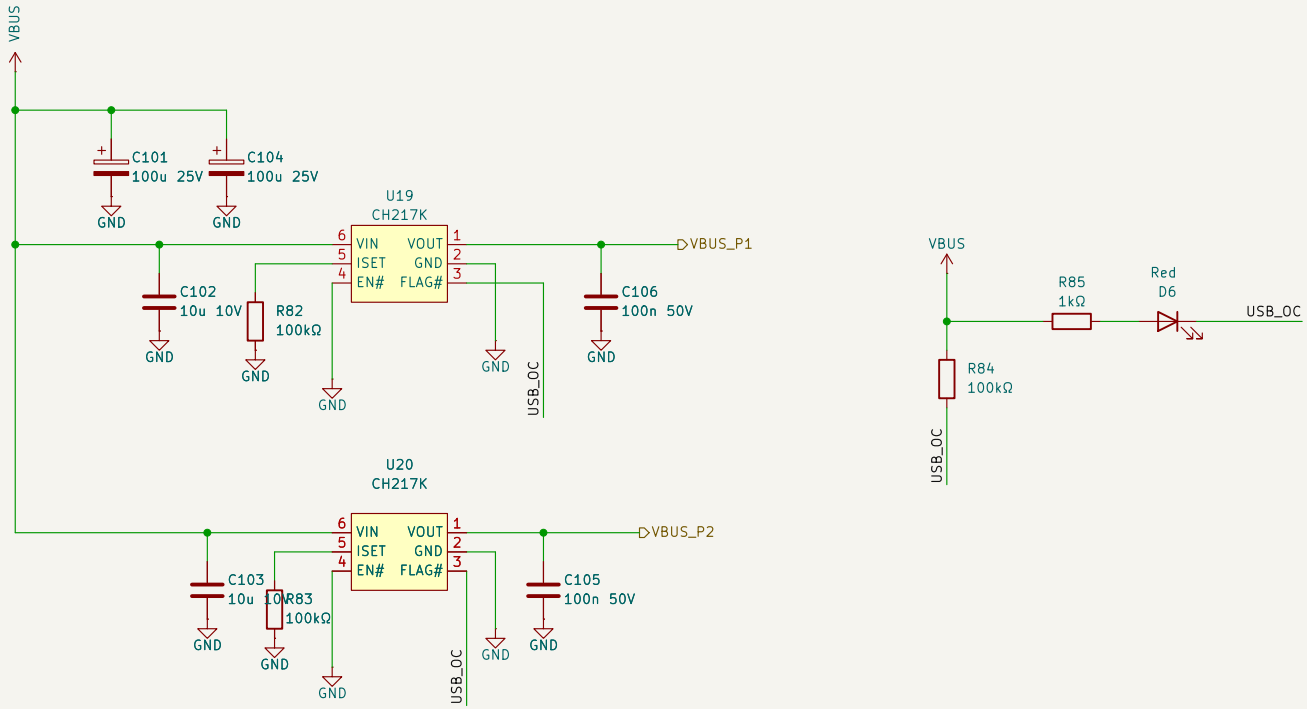
https://github.com/rh1tech/frank

Mikhail Matveev

Sheet: /USB, Power/3V3 Buck/
File: buck3v3.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 17/16



https://github.com/rh1tech/frank

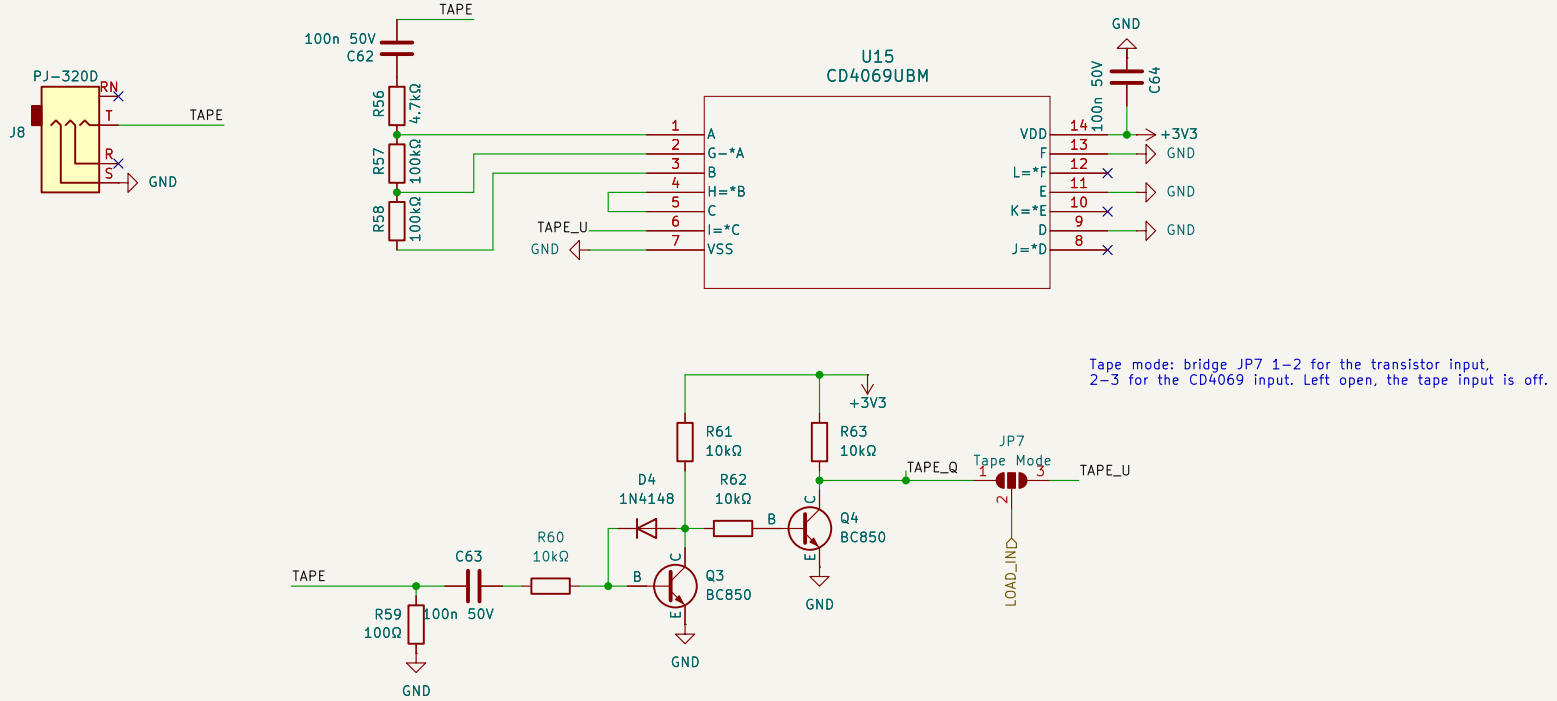
Mikhail Matveev

Sheet: /USB, Power/USB-A Port Power/

File: usb_a_power.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 18/16



https://github.com/rh1tech/frank

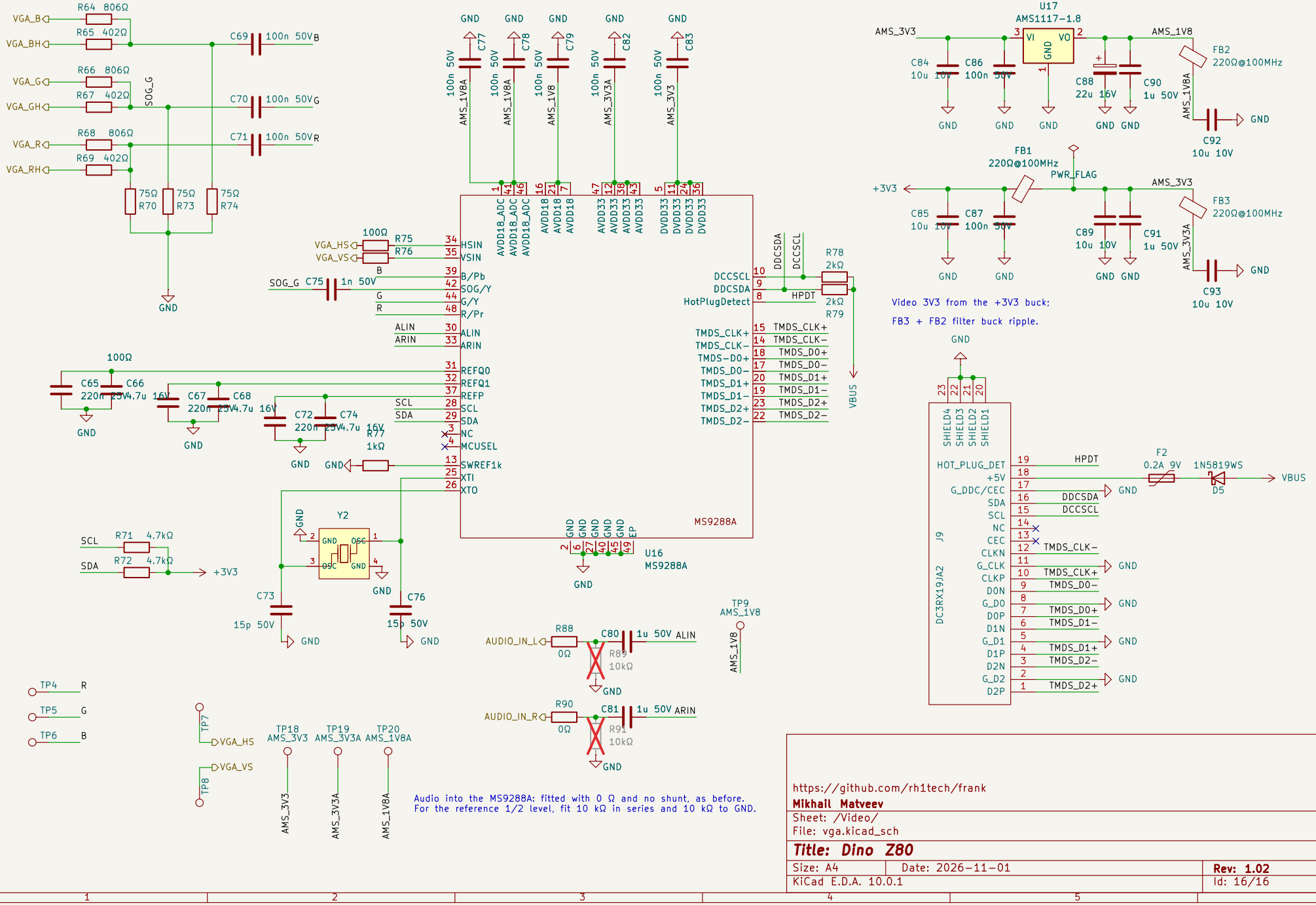
Mikhail Matveev

Sheet: /Tape/

File: tape.kicad_sch

Title: Dino Z80

Size: A4	Date: 2026-11-01	**Rev: 1.02**
KiCad E.D.A. 10.0.1		Id: 15/16


VGA_B
VGA_BH
VGA_G
VGA_GH
VGA_R
VGA_RH
R64 806Ω
R65 402Ω
R66 806Ω
R67 402Ω
R68 806Ω
R69 402Ω
SOG_G
75Ω R70
75Ω R73
75Ω R74
C69 100n 50V B
C70 100n 50V G
C71 100n 50V R
GND
C77 100n 50V AMS_1V8A
C78 100n 50V AMS_1V8A
C79 100n 50V AMS_1V8
C82 100n 50V AMS_3V3A
C83 100n 50V AMS_3V3
AVDD18_ADC
AVDD18_ADC
AVDD18_ADC
AVDD18
AVDD18
AVDD18
AVDD33
AVDD33
AVDD33
AVDD33
DVDD33
DVDD33
DVDD33
DVDD33
100Ω
VGA_HS
VGA_VS
R75
R76
HSIN
VSIN
SOG_G
C75 1n 50V
B/Pb
SOG/Y
G/Y
R/Pr
ALIN
ARIN
REFQ0
REFQ1
REFP
SCL
SDA
NC
MCUSEL
SWREF1k
XTI
XTO
C65 220n
C66 4.7u 16V
C67 220n
C68 4.7u 16V
C72 220n
C74 4.7u 16V
R77 1kΩ
DCCSCL
DDCSDA
HotPlugDetect
R78 2kΩ
R79 2kΩ
HPDT
VBUS
TMDS_CLK+
TMDS_CLK−
TMDS_D0+
TMDS_D0−
TMDS_D1+
TMDS_D1−
TMDS_D2+
TMDS_D2−
MS9288A
U16
Y2
SCL
SDA
R71 4.7kΩ
R72 4.7kΩ
+3V3
C73 15p 50V
C76 15p 50V
AUDIO_IN_L
AUDIO_IN_R
R88 0Ω
R89 10kΩ
R90 0Ω
R91 10kΩ
C80 1u 50V ALIN
C81 1u 50V ARIN
TP9 AMS_1V8
TP4 R
TP5 G
TP6 B
TP7 VGA_HS
TP8 VGA_VS
TP18 AMS_3V3
TP19 AMS_3V3A
TP20 AMS_1V8A
Audio into the MS9288A: fitted with 0 Ω and no shunt, as before.
For the reference 1/2 level, fit 10 kΩ in series and 10 kΩ to GND.
U17
AMS1117-1.8
AMS_3V3
AMS_1V8
C84 10u 10V
C86 100n 50V
C88 22u 16V
C90 1u 50V
FB2 220Ω@100MHz
C92 10u 10V
FB1 220Ω@100MHz
PWR_FLAG
+3V3
C85 10u 10V
C87 100n 50V
C89 10u 10V
C91 1u 50V
FB3 220Ω@100MHz
C93 10u 10V
Video 3V3 from the +3V3 buck;
FB3 + FB2 filter buck ripple.
J9
DC3RX19JA2
SHIELD4
SHIELD3
SHIELD2
SHIELD1
HOT_PLUG_DET
+5V
G_DDC/CEC
SDA
SCL
NC
CEC
CLKN
G_CLK
CLKP
D0N
G_D0
D0P
D1N
G_D1
D1P
D2N
G_D2
D2P
F2 0.2A 9V
1N5819WS
D5
https://github.com/rh1tech/frank
Mikhail Matveev
Sheet: /Video/
File: vga.kicad_sch
Title: Dino Z80
Size: A4
Date: 2026-11-01
KiCad E.D.A. 10.0.1
Rev: 1.02
Id: 16/16